Gene Set Enrichment Analysis (GSEA): Whole Body Hyperthermia (WBH) Treatment

Victoria Chen, Anisha Kolambe, Ethan Aidam

Presented Monday, May 11

Contents

Overview

This analysis starts from the STAR-derived raw gene count matrix (`wbh_counts_matrix.csv`) and walks through quality filtering, differential expression with DESeq2, and a custom implementation of Gene Set Enrichment Analysis (GSEA). We test four biologically motivated gene sets, heat shock/proteostasis, interferon signaling, serotonin, and dopamine, for coordinated expression changes between Whole Body Hyperthermia (WBH)-treated and sham-treated samples.

Load libraries

```
library(AnnotationDbi)
library(org.Hs.eg.db)
library(GO.db)
library(edgeR)
```

```r
library(DESeq2)
library(tidyverse)
library(gt)
```

Load raw STAR count matrix

Column names are cleaned of BAM and tab-file suffixes added by STAR, and Ensembl gene IDs are mapped to HGNC symbols using `org.Hs.eg.db`. Genes with no matching symbol retain their Ensembl ID, and all row names are made unique before downstream use.

```r
cnt <- read.csv(
  "wbh_counts_matrix.csv",
  row.names = 1,
  check.names = FALSE
)
cnt <- as.matrix(cnt)

# Clean sample names if STAR/BAM suffixes are present.
colnames(cnt) <- basename(colnames(cnt))
colnames(cnt) <- sub("\\.Aligned\\.sortedByCoord\\.out\\.bam$", "", colnames(cnt))
colnames(cnt) <- sub("\\.ReadsPerGene\\.out\\.tab$", "", colnames(cnt))

# Remove version numbers if present (e.g., ENSG... .1)
ensembl_ids <- sub("\\..*$", "", rownames(cnt))

gene_annot <- AnnotationDbi::select(
  org.Hs.eg.db,
  keys = ensembl_ids,
  columns = c("SYMBOL"),
  keytype = "ENSEMBL"
)

# Match annotation back to count matrix
gene_symbols <- gene_annot$SYMBOL[match(ensembl_ids, gene_annot$ENSEMBL)]

# Replace missing symbols with Ensembl IDs
gene_symbols[is.na(gene_symbols)] <- ensembl_ids[is.na(gene_symbols)]

# Ensure uniqueness (required for DESeq2)
gene_symbols <- make.unique(gene_symbols)
rownames(cnt) <- gene_symbols

# Sanity checks
if (any(is.na(cnt))) {
  stop("NA values detected in count matrix.")
}
if (any(cnt < 0)) {
  stop("Negative values detected. DESeq2 requires non-negative raw counts.")
}

# Inspect dimensions and first columns
dim(cnt)
```

```
## [1] 78691    19
```

```
head(cnt[, 1:min(4, ncol(cnt))])
```

```
##                 SRR34413163 SRR34413164 SRR34413165 SRR34413166
## DDX11L16                  0           0           0           3
## ENSG00000223972           0           0           0           0
## WASH7P                   53          29          19          11
## ENSG00000227232           0           0           0           0
## MIR6859-1                 0           0           0           0
## ENSG00000243485           3           4           5           4
```

Define Sham versus WBH groups

Each of the 19 samples are assigned to either the WBH group (10 samples) or the Sham group (9 samples) using a manually curated sample table keyed by SRR accession. `Sham` is set as the reference level throughout. A design matrix with an intercept and a `groupWBH` indicator is created for modeling.

```
sample_info <- tibble(
  sample = c(
    "SRR34413163", "SRR34413164", "SRR34413165", "SRR34413166", "SRR34413167",
    "SRR34413168", "SRR34413169", "SRR34413170", "SRR34413171", "SRR34413172",
    "SRR34413174", "SRR34413175", "SRR34413176", "SRR34413177",
    "SRR34413178", "SRR34413179", "SRR34413180", "SRR34413181", "SRR34413182"
  ),
  group = c(
    rep("WBH", 10),
    rep("Sham", 9)
  )
) %>%
  mutate(group = factor(group, levels = c("Sham", "WBH"))) %>%
  as.data.frame()

rownames(sample_info) <- sample_info$sample

# Match count columns to sample metadata.
missing_samples <- setdiff(sample_info$sample, colnames(cnt))
extra_samples   <- setdiff(colnames(cnt), sample_info$sample)

if (length(missing_samples) > 0) {
  stop(paste("Samples in sample_info but missing from cnt:",
             paste(missing_samples, collapse = ", ")))
}
if (length(extra_samples) > 0) {
  message("Samples in cnt but not sample_info will be dropped: ",
          paste(extra_samples, collapse = ", "))
}

cnt <- cnt[, sample_info$sample]

# create a group level vector
group  <- sample_info$group
```

```r
# create design matrix
design <- model.matrix(~ group)

group
```

```
##  [1] WBH  WBH  WBH  WBH  WBH  WBH  WBH  WBH  WBH  WBH  Sham Sham Sham Sham Sham
## [16] Sham Sham Sham Sham
## Levels: Sham WBH
```

```r
design
```

```
##    (Intercept) groupWBH
## 1            1        1
## 2            1        1
## 3            1        1
## 4            1        1
## 5            1        1
## 6            1        1
## 7            1        1
## 8            1        1
## 9            1        1
## 10           1        1
## 11           1        0
## 12           1        0
## 13           1        0
## 14           1        0
## 15           1        0
## 16           1        0
## 17           1        0
## 18           1        0
## 19           1        0
## attr(,"assign")
## [1] 0 1
## attr(,"contrasts")
## attr(,"contrasts")$group
## [1] "contr.treatment"
```

Filter genes by CPM

Lowly expressed genes are removed by retaining only those with a count-per-million (CPM) greater than 1 in at least 2 samples. This step eliminates genes that are essentially absent from the data A `DGEList` object is then constructed and TMM normalization factors are computed.

```r
cpm_vals     <- cpm(cnt)
keep         <- rowSums(cpm_vals > 1) >= 2
cnt_filtered <- cnt[keep, ]

filter_summary <- tibble(
  genes_before_filtering = nrow(cnt),
  genes_after_filtering  = nrow(cnt_filtered),
  genes_removed          = nrow(cnt) - nrow(cnt_filtered)
```

```
)
filter_summary
```

```
## # A tibble: 1 x 3
##   genes_before_filtering genes_after_filtering genes_removed
##                    <int>                 <int>         <int>
## 1                  78691                 25422         53269
```

```
# Store counts and phenotype labels together
dge <- DGEList(
  counts  = cnt_filtered,
  group   = sample_info$group,
  samples = sample_info
)

# Add library sizes and normalization factors
dge <- calcNormFactors(dge)

# Confirm that phenotype labels are stored
dge$samples
```

```
##             group lib.size norm.factors      sample group.1
## SRR34413163   WBH 16335114    1.1235093 SRR34413163     WBH
## SRR34413164   WBH 19804376    0.9208956 SRR34413164     WBH
## SRR34413165   WBH 15374096    1.0706350 SRR34413165     WBH
## SRR34413166   WBH  8878932    1.5650826 SRR34413166     WBH
## SRR34413167   WBH 14379486    0.7819804 SRR34413167     WBH
## SRR34413168   WBH 15811439    1.0181677 SRR34413168     WBH
## SRR34413169   WBH 13107961    0.8028434 SRR34413169     WBH
## SRR34413170   WBH 16466302    1.0409032 SRR34413170     WBH
## SRR34413171   WBH 16202886    0.9948958 SRR34413171     WBH
## SRR34413172   WBH 16100634    1.0074875 SRR34413172     WBH
## SRR34413174  Sham 14400708    1.1965086 SRR34413174    Sham
## SRR34413175  Sham 16732529    0.9654859 SRR34413175    Sham
## SRR34413176  Sham 15296004    0.9503973 SRR34413176    Sham
## SRR34413177  Sham 17257518    0.9549282 SRR34413177    Sham
## SRR34413178  Sham 16627135    1.0604389 SRR34413178    Sham
## SRR34413179  Sham 15043451    1.0610420 SRR34413179    Sham
## SRR34413180  Sham 17403930    0.9380073 SRR34413180    Sham
## SRR34413181  Sham 14386025    0.8961813 SRR34413181    Sham
## SRR34413182  Sham 14048204    0.8721729 SRR34413182    Sham
```

Run DESeq2

DESeq2 is used to fit a negative-binomial model for each gene with `group` as the sole covariate, contrasting WBH against Sham.

```
dds <- DESeqDataSetFromMatrix(
  countData = cnt_filtered,
  colData   = sample_info,
  design    = ~ group
```

```
)

# Ensure Sham is the reference level.
dds$group <- relevel(dds$group, ref = "Sham")
dds       <- DESeq(dds)

res <- results(
  dds,
  contrast = c("group", "WBH", "Sham")
)

summary(res)
```

```
##
## out of 25403 with nonzero total read count
## adjusted p-value < 0.1
## LFC > 0 (up)       : 1, 0.0039%
## LFC < 0 (down)     : 3, 0.012%
## outliers [1]       : 0, 0%
## low counts [2]     : 19, 0.075%
## (mean count < 0)
## [1] see 'cooksCutoff' argument of ?results
## [2] see 'independentFiltering' argument of ?results
```

```
res_df <- as.data.frame(res) %>%
  rownames_to_column("gene_id") %>%
  arrange(padj)

head(res_df)
```

```
##           gene_id   baseMean log2FoldChange      lfcSE      stat       pvalue
## 1           DNM1L  738.35487     -0.2731949 0.05587220 -4.889640 1.010207e-06
## 2           ATG2B 1749.20307     -0.3105544 0.06317638 -4.915673 8.847793e-07
## 3           FKBP4  384.14476      0.7758856 0.16547921  4.688719 2.749202e-06
## 4         GRAMD1C  267.73423     -0.5692414 0.13179513 -4.319138 1.566395e-05
## 5            NCDN  237.75139      0.3571055 0.08446227  4.227989 2.357897e-05
## 6 ENSG00000241860   63.78633      0.7203787 0.17339053  4.154660 3.257715e-05
##         padj
## 1 0.01283115
## 2 0.01283115
## 3 0.02327933
## 4 0.09947783
## 5 0.11979531
## 6 0.13792624
```

Create ranked gene list for GSEA by log2 fold change

GSEA requires a gene list ranked by a metric that captures the magnitude of differential expression. Here we use the DESeq2 log2 fold change (WBH vs. Sham), sorted in decreasing order so that the most upregulated genes appear at the top of the list.

```r
gsea_ranked_log2fc <- res_df %>%
  dplyr::filter(!is.na(log2FoldChange)) %>%
  dplyr::select(gene_id, log2FoldChange) %>%
  dplyr::arrange(desc(log2FoldChange))

head(gsea_ranked_log2fc)
```

```
##           gene_id log2FoldChange
## 1       LINC00958       4.759068
## 2        TBC1D3.1       3.936580
## 3           RAB17       3.200490
## 4 ENSG00000295037       3.027509
## 5       LINC00355       3.017234
## 6         CYP2B7P       2.986241
```

Define gene sets for GSEA from GO IDs

Gene sets are assembled by querying `org.Hs.eg.db` for Biological Process GO terms whose descriptions match four keywords of interest: *heat/chaperone/protein folding/proteostasis*, *interferon*, *serotonin*, and *dopamine*. Only genes present in the ranked list are retained, giving four gene sets of sizes ranging from 46 (serotonin) to 374 (interferon signaling).

```r
# Create named ranked vector for GSEA (gene -> statistic)
ranked_stats        <- gsea_ranked_log2fc$log2FoldChange
names(ranked_stats) <- gsea_ranked_log2fc$gene_id
ranked_stats        <- sort(ranked_stats, decreasing = TRUE)

# Map gene symbols to GO terms (Biological Process only)
go_map <- AnnotationDbi::select(
  org.Hs.eg.db,
  keys    = keys(org.Hs.eg.db, keytype = "SYMBOL"),
  columns = c("SYMBOL", "GO", "ONTOLOGY"),
  keytype = "SYMBOL"
) %>%
  filter(!is.na(GO), ONTOLOGY == "BP")

# Retrieve GO term names (Biological Process ontology)
go_terms <- AnnotationDbi::select(
  GO.db,
  keys    = keys(GO.db),
  columns = c("TERM", "ONTOLOGY"),
  keytype = "GOID"
) %>%
  dplyr::filter(ONTOLOGY == "BP")

# Create mapping: GO ID -> gene symbols
go2gene <- split(go_map$SYMBOL, go_map$GO)

# Identify relevant GO terms by keyword matching
heat_go_ids <- go_terms %>%
  filter(str_detect(
    TERM,
```

```
    regex("heat|chaperone|protein folding|protein refolding|proteostasis",
          ignore_case = TRUE)
  )) %>%
  pull(GOID)

interferon_go_ids <- go_terms %>%
  filter(str_detect(TERM, regex("interferon", ignore_case = TRUE))) %>%
  pull(GOID)

serotonin_go_ids <- go_terms %>%
  filter(str_detect(TERM, regex("serotonin", ignore_case = TRUE))) %>%
  pull(GOID)

dopamine_go_ids <- go_terms %>%
  filter(str_detect(TERM, regex("dopamine", ignore_case = TRUE))) %>%
  pull(GOID)

# Build gene sets from selected GO terms
gene_sets <- list(
  heat_shock_proteostasis = unique(unlist(go2gene[heat_go_ids])),
  interferon_signaling    = unique(unlist(go2gene[interferon_go_ids])),
  serotonin               = unique(unlist(go2gene[serotonin_go_ids])),
  dopamine                = unique(unlist(go2gene[dopamine_go_ids]))
)

# Keep only genes present in the ranked list
gene_sets <- lapply(gene_sets, function(gs) {
  intersect(unique(gs), names(ranked_stats))
})

# Summarize gene set sizes after filtering
gene_set_summary <- tibble(
  gene_set               = names(gene_sets),
  n_genes_in_ranked_list = sapply(gene_sets, length)
)
gene_set_summary
```

```
## # A tibble: 4 x 2
##   gene_set                n_genes_in_ranked_list
##   <chr>                                    <int>
## 1 heat_shock_proteostasis                    293
## 2 interferon_signaling                       374
## 3 serotonin                                   46
## 4 dopamine                                   131
```

Define Enrichment Score function

The enrichment score (ES) is computed using the weighted Kolmogorov–Smirnov-like statistic from the original Subramanian et al. (2005) paper. At each position in the ranked gene list, the running sum increases by the gene's absolute fold-change (weighted hit) if the gene belongs to the set, and decreases by an equal penalty otherwise. The ES is the maximum absolute deviation of this running sum from zero.

```r
gsea_es <- function(ranked_stats, gene_set, p = 1) {
  genes <- names(ranked_stats)
  N     <- length(genes)
  hits  <- genes %in% gene_set
  Nh    <- sum(hits)

  if (Nh == 0) return(NA)

  # Compute weights
  weights <- abs(ranked_stats)^p

  # Normalization constants
  NR <- sum(weights[hits])

  # Initialize cumulative vectors
  Phit  <- numeric(N)
  Pmiss <- numeric(N)

  # Compute cumulative sums explicitly
  hit_cum  <- 0
  miss_cum <- 0

  for (i in seq_len(N)) {
    if (hits[i]) {
      hit_cum  <- hit_cum  + weights[i] / NR
    } else {
      miss_cum <- miss_cum + 1 / (N - Nh)
    }
    Phit[i]  <- hit_cum
    Pmiss[i] <- miss_cum
  }

  running_score <- Phit - Pmiss

  # Enrichment score = max absolute deviation
  ES <- running_score[which.max(abs(running_score))]

  return(list(
    ES            = ES,
    running_score = running_score,
    Phit          = Phit,
    Pmiss         = Pmiss
  ))
}
```

Permutation Test for Normalized Enrichment Score (NES) and P-value

Statistical significance is assessed via phenotype permutation. Rather than refitting the full DESeq2 model 1,000 times (which would be prohibitively slow), we permute sample labels and recompute a fast ranking statistic: the difference in mean log-CPM between the permuted WBH and Sham groups. The observed ES for each gene set is then normalized by the mean of the same-sign null ES values to produce the NES, and an empirical p-value is derived from the fraction of permutations with a null |ES| greater than or equal to the observed |ES|. Multiple testing correction uses both Benjamini–Hochberg (BH) and Storey's q-value.

```r
# phenotype permutation using logCPM mean differences
logcpm <- edgeR::cpm(dge, log = TRUE, prior.count = 1)

rank_from_labels <- function(logcpm, group) {
  group <- factor(group, levels = c("Sham", "WBH"))
  stat  <- rowMeans(logcpm[, group == "WBH",  drop = FALSE]) -
           rowMeans(logcpm[, group == "Sham", drop = FALSE])
  sort(stat, decreasing = TRUE)
}

compute_NES <- function(observed_ES, null_dist) {
  if (is.na(observed_ES)) return(NA_real_)
  if (observed_ES >= 0) {
    pos_null <- null_dist[null_dist >= 0]
    if (length(pos_null) == 0 || mean(pos_null) == 0) return(NA_real_)
    observed_ES / mean(pos_null)
  } else {
    neg_null <- null_dist[null_dist < 0]
    if (length(neg_null) == 0 || mean(abs(neg_null)) == 0) return(NA_real_)
    observed_ES / mean(abs(neg_null))
  }
}

gsea_pheno_perm <- function(dge, gene_sets, observed_ranked_stats, nperm = 1000) {
  logcpm <- edgeR::cpm(dge, log = TRUE, prior.count = 1)

  observed_ES <- sapply(gene_sets, function(gs) {
    gsea_es(observed_ranked_stats, gs)$ES
  })

  null_mat <- matrix(
    NA_real_,
    nrow     = nperm,
    ncol     = length(gene_sets),
    dimnames = list(NULL, names(gene_sets))
  )

  for (b in seq_len(nperm)) {
    perm_group         <- sample(dge$samples$group)
    perm_ranked_stats  <- rank_from_labels(logcpm, perm_group)
    null_mat[b, ]      <- sapply(gene_sets, function(gs) {
      gsea_es(perm_ranked_stats, gs)$ES
    })
  }

  pvalues <- sapply(seq_along(gene_sets), function(j) {
    mean(abs(null_mat[, j]) >= abs(observed_ES[j]), na.rm = TRUE)
  })

  NES <- sapply(seq_along(gene_sets), function(j) {
    compute_NES(
      observed_ES = observed_ES[j],
      null_dist   = null_mat[, j]
```

```
    )
  })

  gsea_df <- tibble(
    gene_set = names(gene_sets),
    ES       = observed_ES,
    NES      = NES,
    pvalue   = pvalues,
    padj_BH  = p.adjust(pvalues, method = "BH")
  ) %>%
    arrange(pvalue)

  lambda    <- 0.5
  pi0_storey <- min(1, mean(gsea_df$pvalue > lambda, na.rm = TRUE) / (1 - lambda))

  gsea_df <- gsea_df %>%
    mutate(
      pi0_storey    = pi0_storey,
      qvalue_storey = pi0_storey * pvalue
    )

  return(gsea_df)
}

set.seed(114)
gsea_df <- gsea_pheno_perm(
  dge                  = dge,
  gene_sets            = gene_sets,
  observed_ranked_stats = ranked_stats,
  nperm                = 1000
)
```

GSEA Results

None of the four gene sets reach statistical significance after multiple-testing correction (all BH-adjusted p-values >= 0.844). The heat shock/proteostasis set shows the most positive NES (1.103), consistent with the expected cellular response to elevated temperature, while the dopamine, serotonin, and interferon sets all have negative NES values, suggesting slight depletion relative to chance.

```
gene_set_labels <- c(
  heat_shock_proteostasis = "Heat Shock Proteasis",
  dopamine                = "Dopamine",
  serotonin               = "Serotonin",
  interferon_signaling    = "Interferon Signaling"
)

gsea_df <- gsea_df %>%
  mutate(
    gene_set = recode(gene_set, !!!gene_set_labels)
  )

gsea_table <- gsea_df |>
```

Gene Set Enrichment Analysis Results

Summary of enrichment scores and significance metrics

Gene Set	ES	NES	P-value	Adj. P (BH)	Pi0 (Storey)	Q-value
Heat Shock Proteasis	0.215	1.103	0.307	0.844	1	0.307
Dopamine	-0.541	-1.017	0.527	0.844	1	0.527
Serotonin	-0.463	-0.958	0.633	0.844	1	0.633
Interferon Signaling	-0.153	-0.638	0.858	0.858	1	0.858

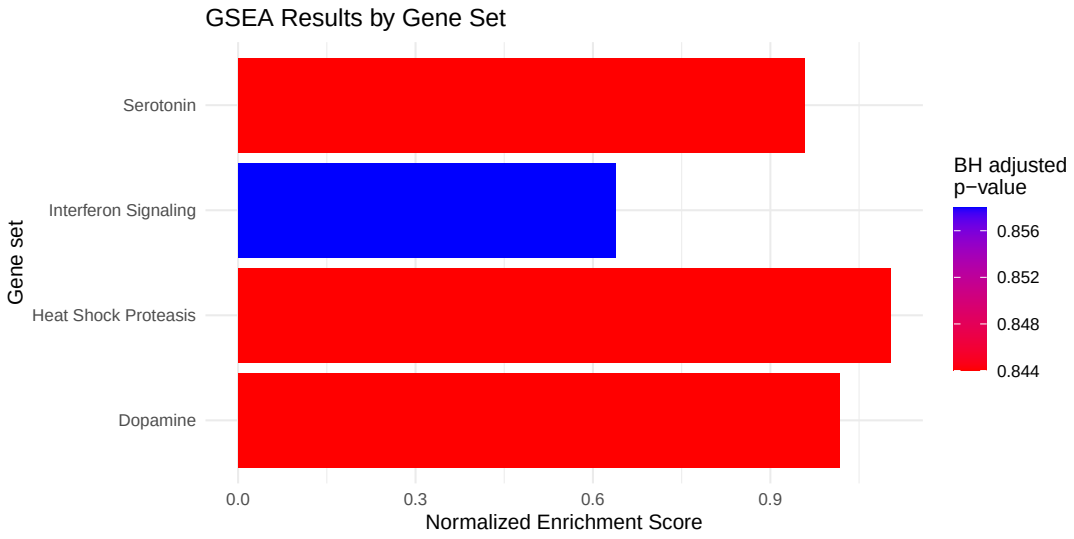
```
  gt() |>
  tab_header(
    title    = "Gene Set Enrichment Analysis Results",
    subtitle = "Summary of enrichment scores and significance metrics"
  ) |>
  cols_label(
    gene_set      = "Gene Set",
    ES            = "ES",
    NES           = "NES",
    pvalue        = "P-value",
    padj_BH       = "Adj. P (BH)",
    pi0_storey    = "Pi0 (Storey)",
    qvalue_storey = "Q-value"
  ) |>
  fmt_number(columns = c(ES, NES), decimals = 3) |>
  fmt_number(columns = c(pvalue, padj_BH, qvalue_storey), decimals = 3) |>
  tab_style(
    style     = cell_text(weight = "bold"),
    locations = cells_column_labels(everything())
  ) |>
  cols_align(align = "center", columns = -gene_set) |>
  tab_options(
    table.font.size     = 10,
    table.width         = pct(100),
    data_row.padding    = px(3),
    column_labels.padding = px(3)
  )

gsea_table
```

Plotting Enrichment Scores

The bar chart below displays the absolute NES for each gene set, colored by its BH-adjusted p-value. Redder bars indicate lower (more significant) adjusted p-values, while bluer bars indicate higher (less significant) values.

```
ggplot(gsea_df, aes(x = gene_set, y = abs(NES), fill = padj_BH)) +
  geom_col() +
  coord_flip() +
  scale_fill_gradient(
    low  = "red",
    high = "blue",
```

```
    name = "BH adjusted\np-value"
  ) +
  labs(
    title = "GSEA Results by Gene Set",
    x     = "Gene set",
    y     = "Normalized Enrichment Score"
  ) +
  theme_minimal()
```



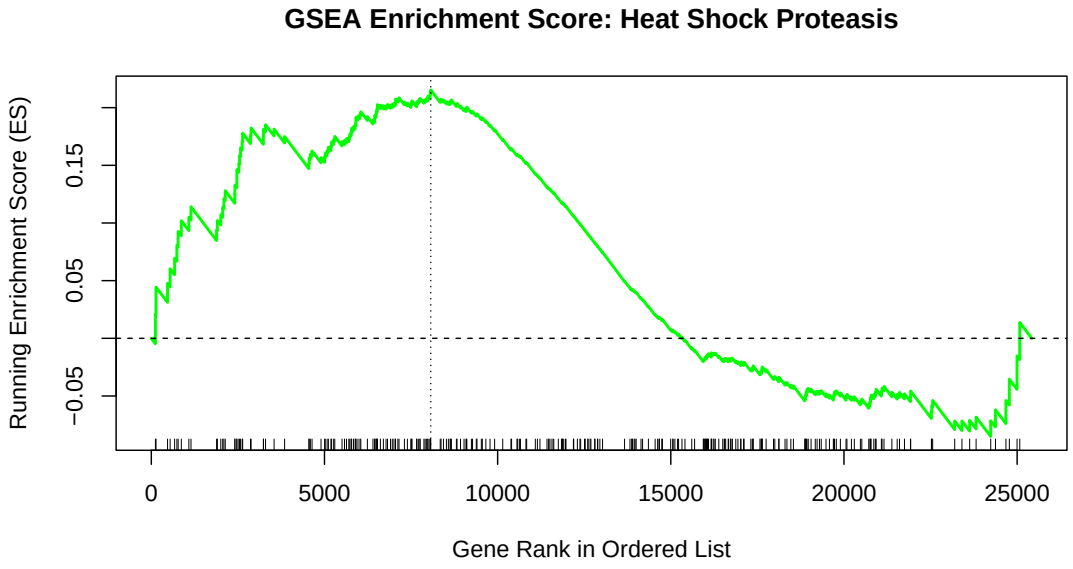
The running enrichment score plots below show, for each gene set, how the cumulative sum evolves as we walk down the ranked gene list from most upregulated (left) to most downregulated (right). Tick marks on the x-axis indicate where gene-set members appear in the list; the dotted vertical line marks the position of the maximum absolute deviation (the reported ES).

```
plot_gsea_es <- function(ranked_stats, gene_set, gene_set_name) {
  es <- gsea_es(ranked_stats, gene_set)

  plot(
    es$running_score,
    type = "l",
    lwd  = 2,
    col  = "green",
    xlab = "Gene Rank in Ordered List",
    ylab = "Running Enrichment Score (ES)",
    main = paste("GSEA Enrichment Score:", gene_set_name)
  )
  abline(h = 0, lty = 2)

  hit_positions <- which(names(ranked_stats) %in% gene_set)
  rug(hit_positions)

  abline(
    v   = which.max(abs(es$running_score)),
    lty = 3
  )
}
```

Heat Shock / Proteostasis

The heat shock gene set (293 genes) shows a broad positive enrichment: the running score climbs steadily through the upper half of the ranked list before falling back, with its peak (ES = 0.215) occurring around rank 8,000. This pattern is consistent with mild but diffuse upregulation of chaperones and proteostasis genes following WBH treatment.

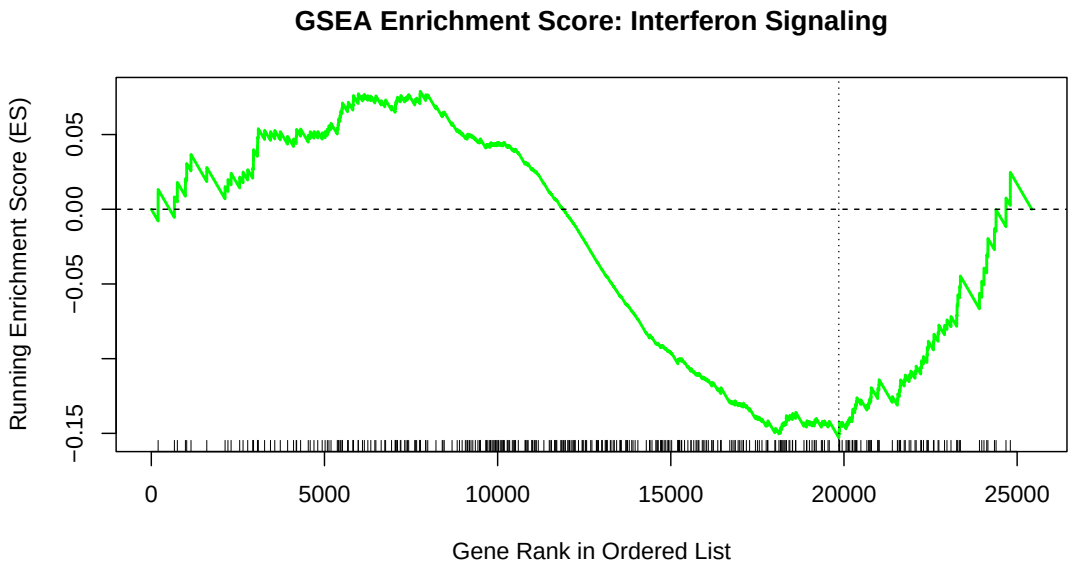
```
plot_gsea_es(
  ranked_stats  = ranked_stats,
  gene_set      = gene_sets[["heat_shock_proteostasis"]],
  gene_set_name = gene_set_labels[["heat_shock_proteostasis"]]
)
```



Interferon Signaling

The interferon signaling set (374 genes) produces a small positive peak early in the list before a sustained decline, yielding a negative final ES (-0.153). This suggests that interferon pathway genes are not preferentially upregulated by WBH.

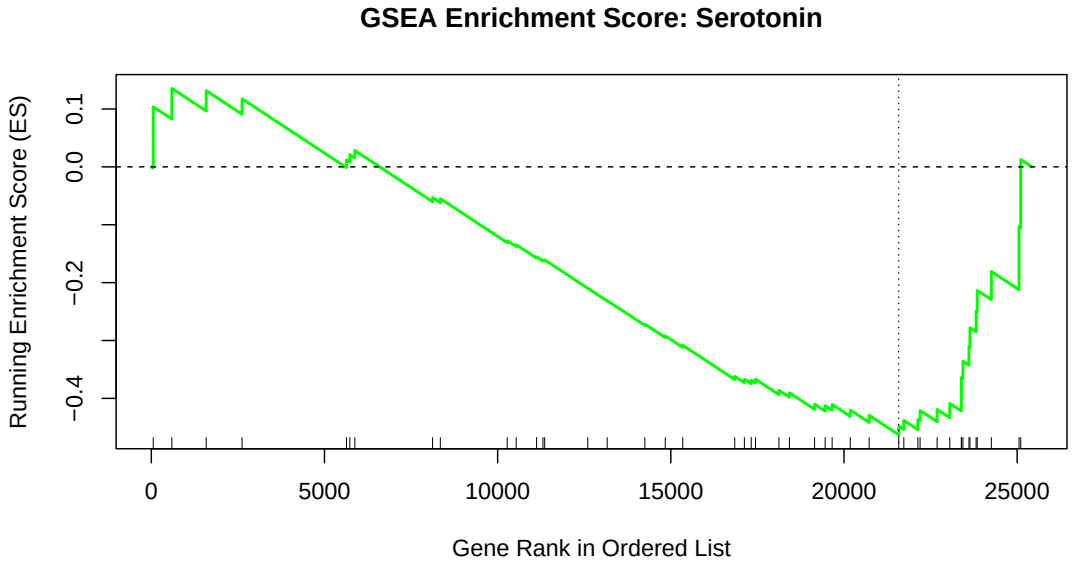
```
plot_gsea_es(
  ranked_stats  = ranked_stats,
  gene_set      = gene_sets[["interferon_signaling"]],
  gene_set_name = gene_set_labels[["interferon_signaling"]]
)
```



Serotonin

The serotonin gene set (46 genes) shows a very early, brief positive peak followed by a nearly linear decline through the remainder of the list, resulting in an ES of -0.463. The plot reflects the small set size and the presence serotonin genes across the lower half of the ranking.

```
plot_gsea_es(
  ranked_stats  = ranked_stats,
  gene_set      = gene_sets[["serotonin"]],
  gene_set_name = gene_set_labels[["serotonin"]]
)
```

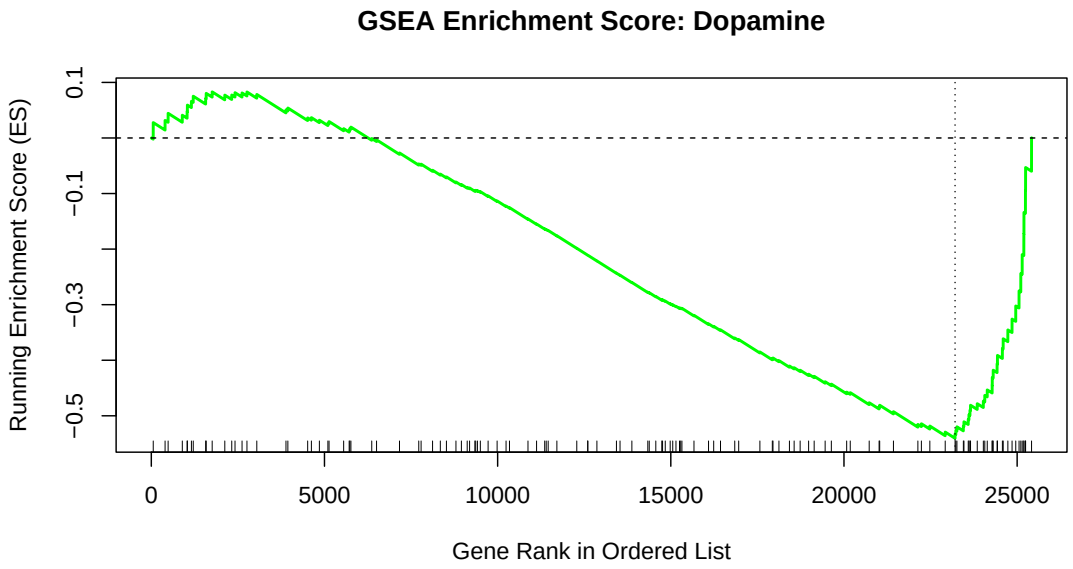


Dopamine

Dopamine pathway genes (131 genes) show a pattern similar to serotonin: a small positive early peak followed by a long linear decline, giving the largest negative ES of the four sets (-0.541). Taken together with the

serotonin result, this suggests that neurotransmitter signaling genes tend to be downregulated, or at least not upregulated under WBH in this dataset.

```
plot_gsea_es(
  ranked_stats  = ranked_stats,
  gene_set      = gene_sets[["dopamine"]],
  gene_set_name = gene_set_labels[["dopamine"]]
)
```



Session info

```
sessionInfo()
```

```
## R version 4.5.2 (2025-10-31)
## Platform: aarch64-apple-darwin20
## Running under: macOS Tahoe 26.3.1
##
## Matrix products: default
## BLAS:   /System/Library/Frameworks/Accelerate.framework/Versions/A/Frameworks/vecLib.framework/Versi
## LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib;  LAPACK v
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: America/New_York
## tzcode source: internal
##
## attached base packages:
## [1] stats4    stats     graphics  grDevices utils     datasets  methods
## [8] base
##
## other attached packages:
##  [1] gt_1.3.0                    lubridate_1.9.5
```

```
##  [3] forcats_1.0.1               stringr_1.6.0
##  [5] dplyr_1.2.1                 purrr_1.2.2
##  [7] readr_2.2.0                 tidyr_1.3.2
##  [9] tibble_3.3.1                ggplot2_4.0.3
## [11] tidyverse_2.0.0             DESeq2_1.50.2
## [13] SummarizedExperiment_1.40.0 MatrixGenerics_1.22.0
## [15] matrixStats_1.5.0           GenomicRanges_1.62.1
## [17] Seqinfo_1.0.0               edgeR_4.8.2
## [19] limma_3.66.0                GO.db_3.22.0
## [21] org.Hs.eg.db_3.22.0         AnnotationDbi_1.72.0
## [23] IRanges_2.44.0              S4Vectors_0.48.1
## [25] Biobase_2.70.0              BiocGenerics_0.56.0
## [27] generics_0.1.4
##
## loaded via a namespace (and not attached):
##  [1] tidyselect_1.2.1    farver_2.1.2        blob_1.3.0
##  [4] Biostrings_2.78.0   S7_0.2.2            fastmap_1.2.0
##  [7] digest_0.6.39       timechange_0.4.0    lifecycle_1.0.5
## [10] statmod_1.5.1       KEGGREST_1.50.0     RSQLite_3.52.0
## [13] magrittr_2.0.5      compiler_4.5.2      rlang_1.2.0
## [16] tools_4.5.2         utf8_1.2.6          yaml_2.3.12
## [19] knitr_1.51          labeling_0.4.3      S4Arrays_1.10.1
## [22] bit_4.6.0           DelayedArray_0.36.1 xml2_1.5.2
## [25] RColorBrewer_1.1-3  abind_1.4-8         BiocParallel_1.44.0
## [28] withr_3.0.2         grid_4.5.2          scales_1.4.0
## [31] tinytex_0.59        cli_3.6.6           rmarkdown_2.31
## [34] crayon_1.5.3        otel_0.2.0          rstudioapi_0.18.0
## [37] httr_1.4.8          tzdb_0.5.0          DBI_1.3.0
## [40] cachem_1.1.0        parallel_4.5.2      XVector_0.50.0
## [43] vctrs_0.7.3         Matrix_1.7-5        hms_1.1.4
## [46] bit64_4.8.0         locfit_1.5-9.12     glue_1.8.1
## [49] codetools_0.2-20    stringi_1.8.7       gtable_0.3.6
## [52] pillar_1.11.1       htmltools_0.5.9     R6_2.6.1
## [55] evaluate_1.0.5      lattice_0.22-9      png_0.1-9
## [58] memoise_2.0.1       Rcpp_1.1.1-1.1      SparseArray_1.10.10
## [61] xfun_0.57           fs_2.1.0            pkgconfig_2.0.3
```